is coherent and rational and enables correct qualitative predictions to be made. It is not for nothing that it remained for centuries the best available model for understanding motion.[1]

Perhaps even more important for the future development of science is Plato.

It is he who understood the value of the intuition of Pythagoras and Pythagorism: that the key to moving forwards, to going beyond Miletus, was mathematics.

Pythagoras was born on Samos, a small island not far from Miletus. His first biographers, Iamblichus and Porphyry, report how young Pythagoras was a disciple of elderly Anaximander. Everything originates in Miletus. Pythagoras travelled widely, probably in Egypt and as far as Babylon, before finally settling in the south of Italy, at Crotone, where he founded a religio-politico-scientific sect which played an important role in the politics of the little town but left a momentous legacy to the entire world: the discovery of the theoretical utility of mathematics: 'Number' – he is said to have asserted – 'governs forms and ideas.'[2]

Plato divested Pythagorism of its cumbersome and useless mystical baggage. He absorbed and distilled its useful message: mathematics is the language best adapted to understand and describe the world. The reach of this insight is immense; it is one of the reasons for the success of Western science. According to tradition, Plato carved on the door of his school the phrase, 'Let no one enter here who is ignorant of geometry.'

Driven by this conviction, it was Plato who posed the momentous question: the question out of which, after a long detour, modern science would emerge. Of his disciples who studied mathematics he asked if they could find the mathematical laws followed by the celestial bodies visible in the heavens. Venus, Mars and Jupiter can be easily observed in the night sky. They seem to move a little at random, back and forth among the other stars. Is it possible to find a mathematics which is able to describe and predict their movements?

The exercise begins with Eudoxus in Plato's school, and is pursued throughout the following centuries by astronomers such as Aristarchus and Hipparchus, bringing ancient astronomy on to an extremely high scientific level. We know of the triumphs of this science thanks to a single book, the only one to have survived: the *Almagest* of Ptolemy. Ptolemy was an astronomer who lived in Alexandria in the first century of our era, under the Roman Empire, when science was already in decline and about to disappear altogether, overwhelmed by the collapse of the Hellenistic world and suffocated by the Christianization of the empire.

Ptolemy's book is a major work of science. Rigorous, precise, complex, it presents a mathematical system of astronomy capable of predicting the